

Designing a System for Data Processing and Visualization in the Public Health System

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Abstract— The paper presents the need to develop a system for processing data from a population opinion survey on immunization. At the beginning, the need and benefits of conducting a survey, the existing ways and methods of implementation are described. The current implementation of the developed system is presented in connection with the implementation of the project "Investigation of immunization coverage and public attitudes during a pandemic" of the University of Ruse. At the end of the paper, the future development of the project is described, as well as the identified problems at the moment. As a conclusion, the need and benefits of using such a system to improve public health are summarized.

Keywords— data, survey, public health system, immunization, COVID-19.

I. INTRODUCTION

The pandemic spread of COVID-19 over the past two and a half years has had an impact on the social and working life of the population. This puts the modern world in extreme conditions and the need to take quick, decisive and dynamically changing measures that require global containment of this infection and its spread [5]. One of the negative consequences is the reduction of people's trust in state institutions and in particular in the health system. A major factor influencing this problem is the spread of a significant amount of "fake news". The ability to distinguish truth from falsehood is knowledge that people acquire through experience and age [3]. Due to the way the judicial system functions and the organization of the media environment, the average consumer of health services can easily become a victim of this kind of manipulation. The latest statistics point to Bulgaria as the EU country with the lowest percentage of adults with completed vaccination status [2]. The result may be refusal to participate in the mandatory immunization schedule, leading to epidemics of vaccine-preventable diseases. A solution to the described problem is a good information policy on the part of the health authorities, aimed at vulnerable groups of the population. This foresees conducting a survey among the population regarding their attitudes towards immunization and health policy, which leads to the need to process large volumes of data. For a qualitative analysis, it is also necessary to compare the new information with previous surveys. Manual processing of this data would increase the execution time and the end result could be inefficient and even meaningless. In order to solve the described problem and in connection with the implementation of activities under the project "Investigation of immunization coverage and public attitudes during a pandemic" of the University of Ruse - Faculty of "Public Health and Healthcare", it is planned to develop and implement a system

for processing data collected by surveying individuals about their views on immunizations.

II. A WAY TO CONDUCT A SURVEY AMONG THE POPULATION

Polling the population about their opinion on important issues can be done in a variety of ways and techniques. One of the most common methods in the recent past is that in which an interviewer physically visits the homes, workplaces, or other places of the respondents. The situation caused by the spread of Covid-19 makes this method inapplicable, since one of the measures taken to deal with the problem is the observance of physical distance. Conducting polls using modern communication methods can overcome this obstacle. For example, a web-based system could be used to send a link to the survey to a specific group of people. They can fill it in and their answers can then be processed and analyzed. One of the problems that can arise is that the filling can be from a third party, and not from the one whose opinion we are interested in. Various methods can be applied to deal with this problem, which complicate the polling process. An example of a system that can be used to gather information about the opinion of a large group of individuals on a set of questions is Google Forms. Through it, you can ask specific questions, mark some of them as mandatory, and offer a list of acceptable answers. It is also possible to pre-validate the response for eligibility. The use of such modern polling systems by government bodies would be well received by teenagers as they use modern information technology massively and thus their opinion will be counted and researched.

III. OVERVIEW OF EXISTING SOLUTIONS

The program "Statistica" is one of the most common and well-known statistical data processing and analysis systems. This is due to the fact that in many schools and universities students are trained with its functional capabilities. "Statistica" can be defined as software designed for data analysis. They can be extracted, processed, visualized, and there is also the possibility of machine learning. In addition to diagrammatic representation, visualization can be accomplished by the presence or absence of a value from given sets, represented by differently colored dots. An advantage is the universality of the data used and the availability of many tools for their processing. The main disadvantage is the need for the person working with it to have prior training to work with the system, which may take some time.

Another commonly used method is to use an office program to process tables or text. An example of such products are MS Office and Libre Office. The user has the

option to load a file in .csv format. These types of files are generated by survey systems such as Google Forms. Data can be processed and visualized using charts. An advantage is the distribution of these products among the mass consumer, but the main disadvantage is the considerable time required to create a document, in the content and scope of which the processing will take place. A problem can arise when it is required to analyze how public opinion has changed on an issue over a period of time. To do this, it is necessary to compare data from different tables, which may have been created using a different version of the office suite and saved in a different format. The user may need to convert the data from one survey to another format, causing some losses that he may not be aware of. In this way, a distortion in the final result can be obtained. In the past, macros were often used to automate some of the calculations and data processing. In today's times, their use is not recommended due to the fact that the protection of the system is lowered and it becomes vulnerable to malicious persons aiming for a cyber-attack.

IV. DESIGNING THE SYSTEM

After the analysis of the existing solutions and in connection with the implementation of the tasks set under the project "Investigation of immunization coverage and public attitudes during a pandemic" it is necessary to create a software product that meets the following conditions:

- Able to read data from a .csv file.
- Visualized the read data using diagrams.
- Compare the data from a survey conducted in a different period.
- Offer an easy-to-use interface.

One of the tasks of the project includes the need to conduct a poll among the population in relation to attitudes about vaccination. People must belong to different groups formed by social affiliation, level of education, ways of informing about health protection methods and techniques. In the context of a pandemic and post-pandemic situation, the most acceptable and effective way to do this is to use an online survey implemented through the functionality of the Google Forms system. After participants complete the survey, the results can be saved to a .csv file. A CSV (comma separated values) [1] file is a data file where individual values are separated by commas. The file generated by Google Forms has the following structure – the first line contains the survey questions, and each subsequent line contains their answers. If a respondent did not answer a certain question, the answer is a space separated by commas. After reading the data, for easier analysis and perception, it is necessary to present them by means of a diagram, with individual values in percentages. One of the tasks of the project is to determine public attitudes about immunizations. A good option is to compare how people's opinion has changed before, during and after a pandemic situation, which requires the system to be able to compare several polls conducted at different times. The analysis and work with the data will be carried out by medical professionals, which requires the system to

offer a suitable and easy-to-use interface, so as not to commit the workers' time to learning the system.

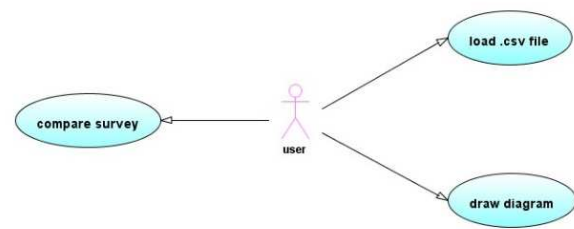


Figure 1. Use case diagram

Figure 1 presents the use cases of the future program. It shows the program functionality, which is to load a .csv file, draw a chart and compare two or more surveys.

V. CURRENT IMPLEMENTATION

As of the current implementation moment of the program, the loading of a .csv file and the drawing of charts for specifically selected data have been completed. The programming language used is java. The jfreechart library is used to render the charts.

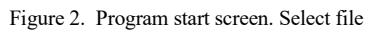
After displaying the main window, the user must specify the path to the file. The work algorithm for visualizing the information in tabular form can be described by the following sequence:

- Open the files.
- Each line is read character by character until a comma character is reached, with the characters up to the first comma, those between two commas and those from the last comma to the end of the line represented as separate array elements.
- <line number, array> structure is created, in which the information about each read line from the .csv file is stored.
- Index 0 of the array stores the data of the list containing the survey questions, and the following indices contain the answers to the questions.
- Visualize the information in tabular form, the data from the list with index 0 is used as column header. A cycle follows that generates a new row from the table until the size of the list is reached. These lines contain the answers to the questions.

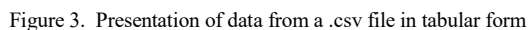
The algorithm for visualizing the data through charts includes the following steps:

- Select a column number from the table and using a loop create a new list containing all elements with this index from the general data list (the first column of the table is numbered 0).
- The first element of the list contains the question and the remaining elements are its answers. As chart titles, an element with index 0 is retrieved and then deleted. The remaining elements of the list are converted into a Map structure, where the key is the answer, and the value is how many times it is repeated in the list. Data

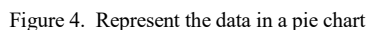
we will be able to analyze how public attitudes have changed on the matching questions.



An important and necessary function is the ability to be able to examine the dependence of a certain factor on a given response. For example, to be able to present a reference in relation to the age of the interviewees, what is their opinion regarding the placement of vaccines from the mandatory immunization calendar. A person may answer the questions without expressing their true opinion, i.e. to answer that they would get vaccinated against COVID-19, but on a subsequent question to mark that vaccines are produced for profit. This would cause the result to be skewed and lead to the wrong measures being taken in relation to a particular situation. Certain control questions must be included in the surveys, on the basis of which it will be possible to filter out incorrectly completed surveys. The program will automatically perform this analysis and display a message with the compromised ones, removing them from the final results. This automated pre-processing would save valuable time for the person involved in the data analysis.



Each person digests the information presented to them in a different way. Therefore, a program must be able to visualize the data in different forms. The presentation of the data, apart from a pie chart, should be expanded using other popular formats: histogram, dot plot, relative freq histogram, stem and leaf display, bar graph and others. A given piece of information can be presented clearly by means of a specific diagram. For example, the division by age of the respondents whether a positive answer about vaccines will be presented better by means of a bar diagram than by means of a stem and leaf display.



Performing a qualitative analysis of a particular survey requires knowing how user opinion has changed on a given issue over a period of time. Comparing data from a survey completed by a group of people at different intervals provides a solution to the problem. It is planned to add a functional possibility for simultaneous loading of several .scv files to the current system. The questions contained in the individual files will be compared for uniformity. If in one survey they follow the same order, but in others the order is reversed, this will not be a problem. In the output, they will be sorted. If a particular survey contains more questions or there are differences, only those that match will be taken. For example, if we have conducted three polls in the last five years, but in the last one, some of the questions were dropped and new ones were added,

Figure 5 presents the program functionality. It shows that when the user loads data files, question matching checks are done automatically and the user does not need to take any additional action. Differences will be removed automatically and he will be able to view the processed information tabularly or by means of diagrams.

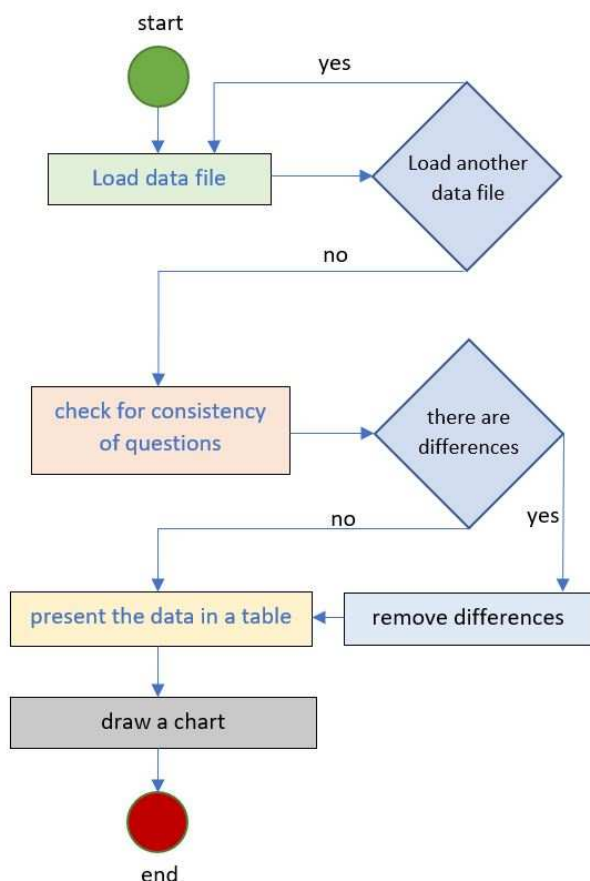


Figure 5. Program functionality

VII. CONCLUSION

During an emergency epidemiological or pandemic situation, the burden on the health system increases many times over, and the health authorities must, in a short period of time, take the most appropriate policies to deal with the problem. A range of measures has also been suggested which necessitated social exclusion, distance learning, limiting contacts, wearing protective equipment, using disinfectants, lockdown of the countries around the world, and all this aiming at stopping the transmission and prevent the spread of the pandemic as rapidly as possible [6]. COVID-19 is causing an overwhelming psychosocial impact on all of humanity [4]. Knowing the public attitudes and opinion of the population, the solution to the problem and the measures taken will find easier applicability and effectiveness. Large amounts of information need to be processed in a minimal amount of time. The persons who are directly interested in the result in most cases are medical professional. Under the project: "Investigation of immunization coverage and public attitudes during a pandemic", a system is being developed to facilitate the collection of survey data among the population, as well as their subsequent processing. For testing purposes, a small survey was conducted among a representative sample of the population, asking fifteen questions in total. The survey was conducted using the Google Forms platform. The results are saved in a .csv file. The part of the system implemented so far successfully reads the data and visualizes it in tabular form. The answers to the individual questions can also be presented

using a pie chart. Working with the program does not require specific knowledge and anyone with basic computer skills would be able to do it successfully. The implementation of the planned future improvements and addition of new functionalities would not complicate the operation of the software.

Conducting a survey using the Google Forms platform would cover a large number of people. Distribution of the entry link may be done through social media, e-mail or media announcements. The study is carried out without physical contact, which during a pandemic emergency situation leads to the fulfillment of the condition of physical distance. The data is easily exported to a .csv file and can be loaded into the developed system, which will present it in a convenient form, as well as perform certain processing. Future validation and remediation of compromised data will contribute to the proper taking of measures and actions to address the problem. The processing can also be done using various universal and multifunctional systems such as Statistica or SPSS, but the user must spend considerable time learning to work with the program.

As a conclusion, it can be stated that the implemented part of the system, as well as the planned future implementation, successfully meets the set goals of project implementation: "Investigation of immunization coverage and public attitudes during a pandemic".

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